

AEROSP-588-Project

Propeller Blade Optimization

Naman Makkar (nmakkar@umich.edu), Vishwa Mehta (mvishwa@umich.edu)

December 2023

1 Abstract

This project is undertaken with the intent of finding the optimum twist distribution across the span of a propeller blade of a fixed-wing military Unmanned Aerial Vehicle having a single propeller on the back. The twist angle throughout the span is optimized to maximize the propeller's efficiency, and to minimize the noise generated by the propeller to help with long range stealth missions at different phases of the mission profile.

The optimization problem is hence a Multi-Objective problem that brings together the disciplines of the Performance Efficiency (η) Model and the Aeroacoustic (Sound Pressure Level) Model to find the optimum twist distribution of the propeller blade. Each propeller blade is discretized into several sections in order to increase the accuracy obtained by the low-fidelity solvers. Both the disciplines are implemented using numerical models that are coupled with a low-fidelity Aerodynamic Solver that solves for the different discretized state-variables that are used by both the disciplines. The combined objective from both the disciplines is then implemented for optimization using Scipy's SLSQP for constrained optimization.

2 Background and Motivation

The increase in the applications for Propeller based Fixed Wing Unmanned Aerial Vehicles in the Defence Industry (like the MQ9 Reaper) has paved way for a huge number of Innovations for the same. Some common problems with propeller based UAVs is their range and the noise generated by them that minimize their scope of operation at lower altitudes where they can be easily spotted and hence limiting their applications in certain scenarios.

These problems have been approached in this project with the aid of Multidisciplinary Design Optimization (MDO), using numerical models to approach the Propeller's Performance Efficiency discipline and the Aero-acoustic discipline for determining the Sound Pressure Level generated by the propeller at an arbitrary distance, and used to determine the optimum twist distribution to maximize the efficiency while minimizing the noise generated with constrained power and thrust requirements for its successful operation.

3 Problem Formulation

3.1 Optimization Problem Statement

$$\text{minimize } SPL_{LF}(\epsilon) + \tau((1 - bC_d) \cos(\phi) - (bC_l + \frac{dC_d}{dC_l}) \sin(\phi)) - \lambda \xi((1 - bC_d) \sin(\phi) + (bC_l + \frac{dC_d}{dC_l}) \cos(\phi))$$

$$\text{w.r.t } \epsilon$$

$$\text{s.t } C_T \geq 0.6$$

$$\text{where } \lambda = \frac{1}{J + 2\pi\lambda_L}$$

Where λ_L is the Lagrange Multiplier published by Lock et al. [1] and is taken to be a constant value of $\lambda_L = 0.04$ for this paper. Where $J = \frac{V_\infty}{\omega D}$ is the advance ratio of the propeller blade.

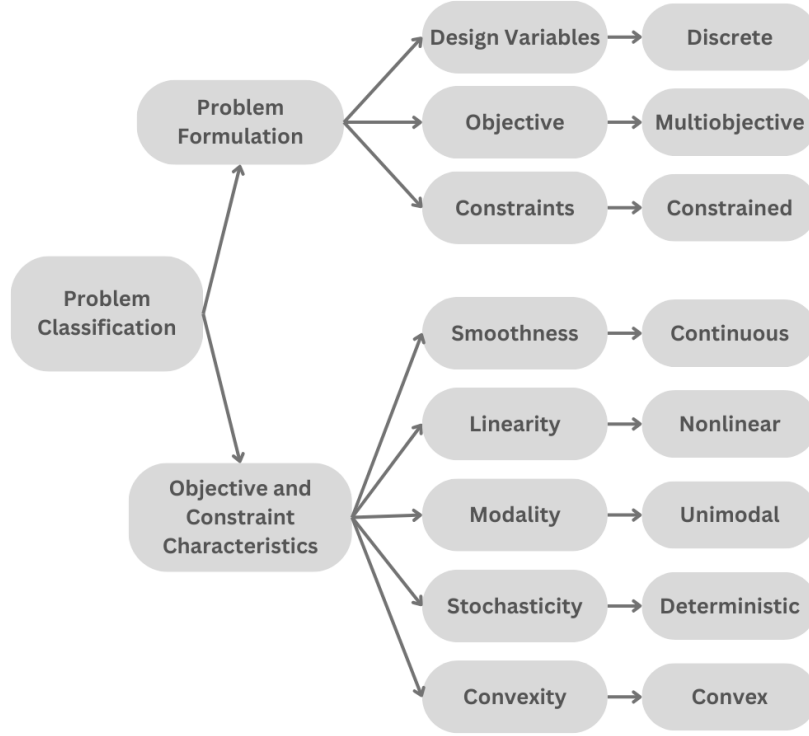


Figure 1: Optimization Problem Classification

3.2 Problem Classification

The final optimization problem can be classified as a coupled, constrained multi-objective problem with discrete design variables. The objective and the constraint functions are continuous, non-linear, convex, deterministic and unimodal. The flow chart for the problem classification can be observed in **Figure 1**

3.3 Description of Models and Coupling

The optimization uses the combined objective obtained from two numerical models namely, the Propeller Performance Efficiency solver, and the Aero-acoustic solver. Both the solvers make use of the various state-variables that are defined by the Aerodynamic Solver. The operation of the optimization occurs as depicted by the **following XDSM plot:**

Aerodynamic Solver:

In order to enhance the accuracy of the low-fidelity models and solvers being used in the project to cut-down upon the computational loads, the entire propeller blade is segmented (discretized) into smaller sections to reduce the scope of approximations on the entire blade, to hence be able to account for the changes in the velocity, lift, drag, thrust and other properties of each section as we move along the span of the blade. The aerodynamic solver, hence provides the several state variables for the current twist distribution along the entire blade of the propeller (at each iteration of the optimization), which are then used by both the solvers to account for the changes and improvement in the value of the combined objective function as the optimization progresses. The NACA2412 airfoil was utilised for modelling the thickness to chord ratio of the propeller blade.

The state-variables include:

Coefficient of Thrust:

$$C_T = \int_{x_{max}}^{x_{min}} T_c dx \quad (1)$$

$$T_c = \tau(C_l \cos \phi - C_d \sin \phi) \quad (2)$$

$$\tau = \frac{\sigma \pi}{8}(J^2 + \pi^2 x^2) \quad (3)$$

Coefficient of Power:

$$C_P = \int_{x_{max}}^{x_{min}} P_c dx \quad (4)$$

$$P_c = \xi(C_l \sin \phi - C_d \cos \phi) \quad (5)$$

$$\xi = \frac{\sigma \pi^2 x}{8}(J^2 + \pi^2 x^2) \quad (6)$$

Coefficient of Lift:

$$C_l = \frac{8xG_0 \sin \phi_0}{\sigma}(\alpha_i) \quad (7)$$

Where the constant G_0 is Goldstein's tip-loss factor [2]

Coefficient of Drag:

$$C_d = 0.1(C_{l,min}C_d - C_l)^4 + C_{d,min} \quad (8)$$

where

$$\frac{C_{lc}}{C_{li}} = \frac{1}{\sqrt{1-M^2}} + \frac{t/c}{1+t/c} \left[\frac{1}{\sqrt{1-M^2}} \left(\frac{1}{\sqrt{1-M^2}} - 1 \right) + \frac{1}{4}(\gamma+1) \left(\frac{1}{1-M^2} - 1 \right)^2 \right] \quad (9)$$

These state variables are generated for each discretized section of the blade and provided to the solvers at each iteration.

Propeller's Performance Efficiency (η) Solver:

This is a numerical model that solves for the efficiency of the propeller along with the value of the Euler-Lagrangian Equation that is utilized to determine the optimality conditions with respect to the propeller's efficiency. The main equations used by this solver are:

The **efficiency** of the propeller is given by the equation:

$$\eta = \frac{J \int_{x_{max}}^{x_{min}} T_c(x, \beta(x)) dx}{\int_{x_{max}}^{x_{min}} P_c(x, \beta(x)) dx} \quad (10)$$

In order to get the optimal distribution of twist, the twist is modified by appending an arbitrary function $\mu(x)$, in product

with a constant named ϵ .

$$\beta_\epsilon(x) = \beta(x) + \epsilon\mu(x) \quad (11)$$

The resultant efficiency of the propeller as a function of epsilon is

$$\eta_\epsilon(\epsilon) = \frac{J \int_{x_{max}}^{x_{min}} T_c(x, \beta_\epsilon) dx}{\int_{x_{max}}^{x_{min}} P_c(x, \beta_\epsilon) dx} \quad (12)$$

In order to get the optimal twist distribution, the efficiency is differentiated by ϵ to obtain the optimum value of ϵ as the other components of the twist function are fixed arbitrarily. Hence, for a particular distribution of $\mu(x)$, the optimal twist is obtained by solving for the optimal value of ϵ . [3]

This gives us:

$$\frac{\delta\eta_\epsilon(\epsilon)}{\delta\epsilon} = \frac{JC_P \int_{x_{max}}^{x_{min}} (\delta T_c / \delta \beta_\epsilon) (\delta \beta_\epsilon / \delta \epsilon) dx - JC_T \int_{x_{max}}^{x_{min}} (\delta P_c / \delta \beta_\epsilon) (\delta \beta_\epsilon / \delta \epsilon) dx}{C_P^2} = 0 \quad (13)$$

This can be further simplified to obtain the unconstrained E-L Equation:

$$C_P \frac{\delta T_c}{\delta \beta} - C_T \frac{\delta P_c}{\delta \beta} = 0 \quad (14)$$

In order to obtain a physically feasible optimum, the power coefficient is constrained to obtain upon simplification:

$$\frac{\delta T_c}{\delta \beta} - \lambda \frac{\delta P_c}{\delta \beta} = 0 \quad (15)$$

Thus, the Euler-Lagrangian Equation allows us to implicitly constrain the power coefficient by using lambda given by:

$$\lambda = \frac{1}{J + 2\pi\lambda_L} \quad (16)$$

where,

$$J(x) = \frac{V_\infty}{\omega D} \quad (17)$$

The constrained E-L Equation is then simplified to get:

$$\tau((1 - bC_d) \cos(\phi) - (bC_l + \frac{dC_d}{dC_l}) \sin(\phi)) - \lambda \xi((1 - bC_d) \sin(\phi) + (bC_l + \frac{dC_d}{dC_l}) \cos(\phi)) = 0 \quad (18)$$

Aeroacoustic Solver:

Aeroacoustic analysis is carried out with a low fidelity model developed based on the parameters defining the geometry and the operating conditions of the propeller. The model used in this project was derived by Schlegel et al. [4] and is given below as -

$$SPL = 10 \log_{10}(KA_b V^6) + 20 \log_{10}(\frac{C_l}{0.4}) - 20 \log_{10}(\frac{\Delta S}{\Delta S_{ref}}) \quad (19)$$

Where $K = 6.11 \cdot 10^{-11} s^6 / ft^8$ is a constant, V is the velocity of the blade at location $x = \frac{r}{R}$ on the blade and, C_l is the coefficient of lift at the same location. $\Delta S_{ref} = 300ft$ is the reference length used by Schlegel and ΔS is the distance from the drone to the observer, which is taken to be 100 ft for this project. This is equation can be simplified to -

$$SPL = 20 \log_{10}(K \frac{V}{\rho \Delta S} \sqrt{\frac{N_b T^2}{2A_b}}) \quad (20)$$

Where $A_b = c \cdot R$ is the area of the blade and the density ρ is taken to be $1kg/m^3$ for a cruising altitude of 5000 feet. This low-fidelity aeroacoustic solver helps in calculating the sound pressure level in decibels.

3.4 Proposed Optimization Algorithm

The proposed optimization algorithm consists of a multi-objective problem with 2 constraints and coupled variables, where both the objectives are calculated using different solvers. The propeller blade is divided into 10 parts and discretized, giving us a 10-D optimization problem which is optimized using a gradient based algorithm.

4 Investigation on Coupled Models

4.1 Bounds

For optimising the multi-objective function, we put a bound on the coefficient of thrust C_T using the constraint $C_T \geq 0.6$ to ensure that SPL stays positive, since the low-fidelity solver tries to minimize the sound pressure level sometimes converging to a negative solution which is not physically possible.

4.2 Modality

The modality of coupled solvers needs to be investigated for each of the problem formulations - unconstrained Euler-Lagrangian equation optimization, constrained Euler-Lagrangian equation optimization and the multi-objective optimization problem which combines the Euler Lagrangian equation with the SPL solver.

We observed that the unconstrained EL equation and the unconstrained SPL equation were multimodal, and applying the constraints allowed the multiobjective problem to converge to one solution, making it unimodal.

4.3 Numerical Noise

The low-fidelity aeroacoustic solver, aerodynamic solver and propeller efficiency solver made use of algebraic relations and equations, hence the data generated is noiseless since it consists of calculations and provides us with a set of coupling variables. However, the data generated by the solvers was occasionally non-differentiable at the points of discretization.

4.4 Derivative Computation

Finite differences were used for computing derivatives. Scipy's gradient based SLSQP was utilised due to the 10D optimization problem. Due to the high dimensional nature of the problem, gradient based optimizers were preferred to gradient free optimizers. Finite differences were implemented using Scipy's solver. Finite differences had to be utilised due to the fact that the objective function and the variables had to be calculated across the blade by discretization and were not differentiable everywhere.

5 Optimization of a simplified Problem

For the simplified optimization problem, unconstrained and constrained optimization of efficiency using the Euler-Lagrangian equation was carried out utilising the same variables. Unconstrained optimization for the Euler-Lagrangian equation was initially implemented without discretization using a 1-D initial value for ϵ and then the unconstrained and constrained Euler-Lagrangian equations were optimised for a 10-D initial value of ϵ . It was observed that for the unconstrained Euler-Lagrangian optimization in **Figure 3** and **Figure 5**, the optimal distribution obtained for the twist angle β had a significantly larger, when compared to the optimal range for β for the constrained problems of the respective distributions. This however, was not the case for **Figure 4**.

6 Optimization Results

The optimal distribution of twist is obtained for each phase of the mission profile by adjusting the velocity of advance and the air density in accordance with the phase of flight for an altitude of *5000ft*. The result obtained is as follows:

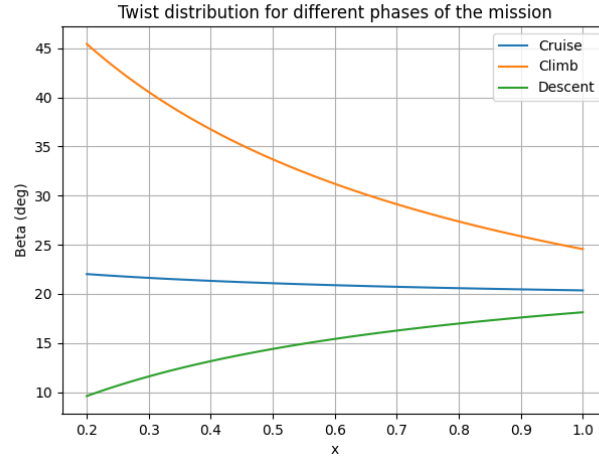
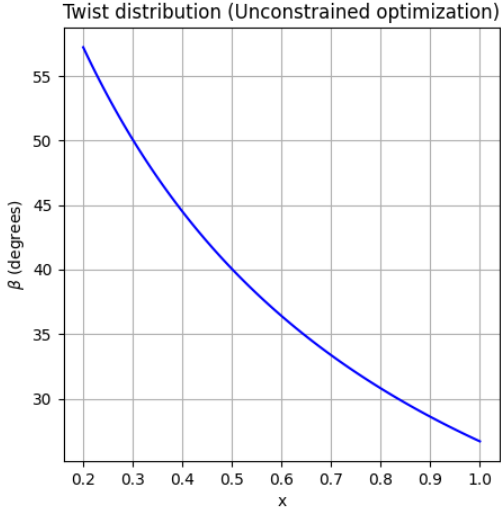
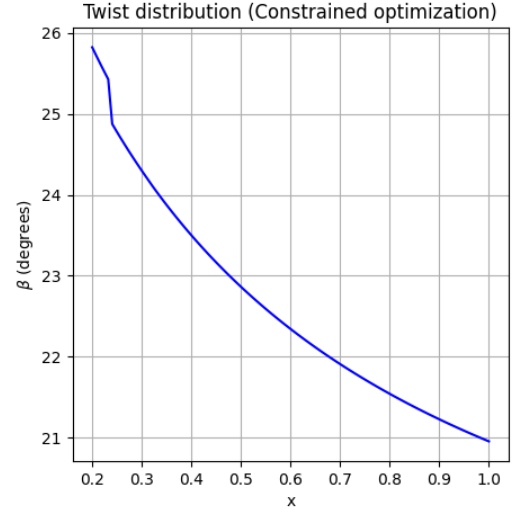


Figure 2: Optimum Twist Distribution along the blade during different mission phases

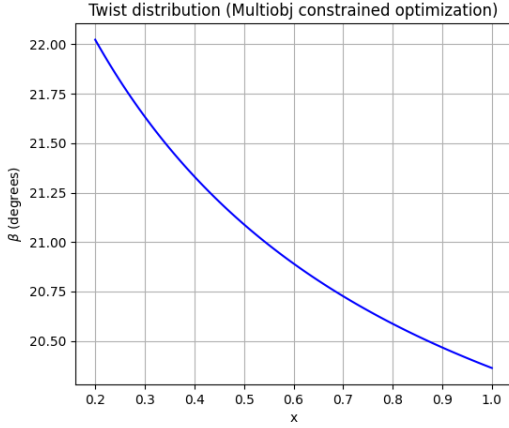
This optimum is obtained with having $\mu(x) = \frac{1}{x-0.5} - 0.5$ (which is the most similar distribution to the one used by [3]). Additionally an in-depth analysis considering various distributions of $\mu(x)$ were performed while varying the number of blades for all 3 optimization problems - unconstrained Euler-Lagrangian optimization, constrained Euler-Lagrangian optimization and the coupled multi-objective optimization problem. The optimal distributions of the twist angle β were calculated for each optimization problem and the results can be observed in **Figure 3-5**:



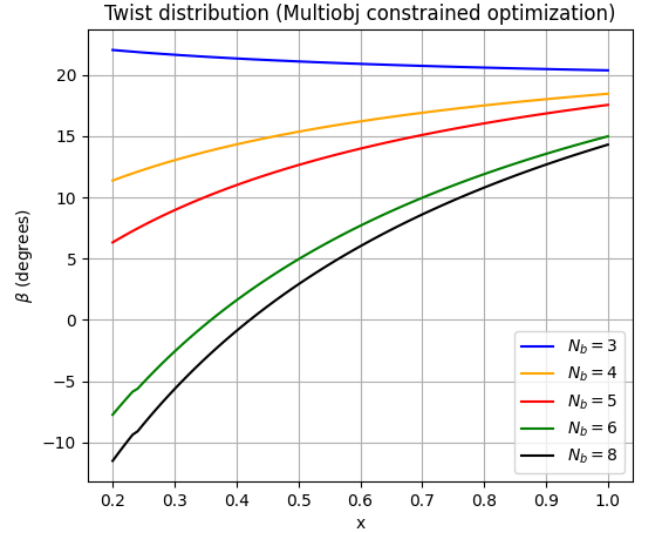
(a) Distribution plot for β vs x for the unconstrained Euler-Lagrangian Objective



(b) Distribution plot for β vs x for the Euler-Lagrangian Objective using Power Constraints

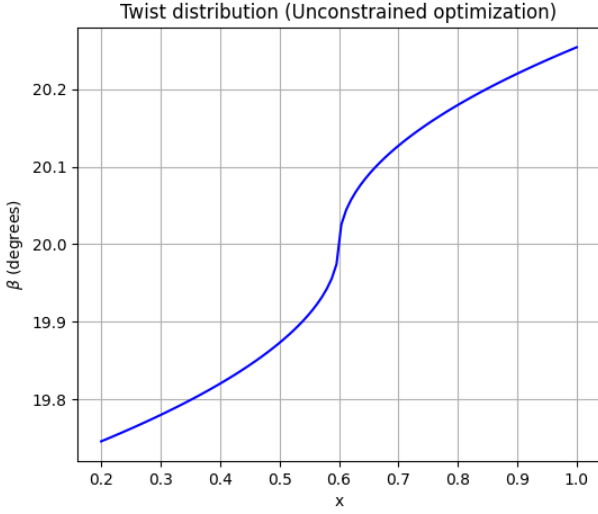


(c) Distribution plot for β vs x for the Multi-objective problem which optimises the Euler-Lagrangian and Sound Pressure Level objectives for thrust and power constraints for $N_b = 3$ blades.

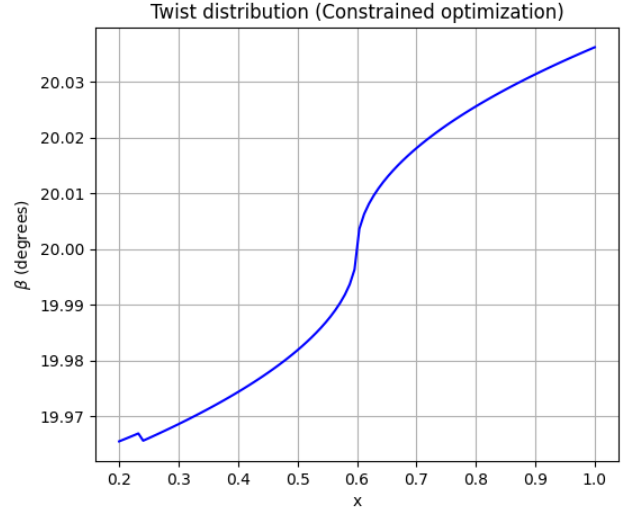


(d) Distribution plot for β vs x for the Multi-objective problem for different number of blades N_b

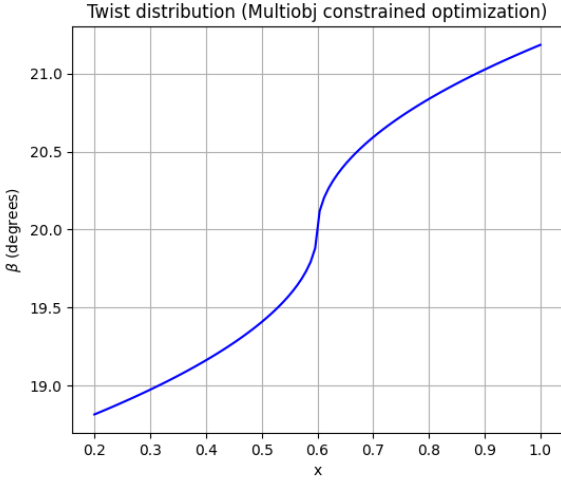
Figure 3: Distribution plots for the twist angle β vs $x = \frac{r}{R}$ for $\mu(x) = \frac{1}{x-0.5} - 0.5$ which is the most similar distribution to the one used by [3]. Experiments were conducted for different number of blades N_b for the propeller, varying from $N_b = 3$ to $N_b = 8$, which can be observed in **Figure d**. It was observed that the distribution obtained by optimising the EL-equation alone did not show any variations for different N_b , however different optimal distributions were observed for the Multi-objective problem due to the fact that the Sound Pressure Level varies with N_b .



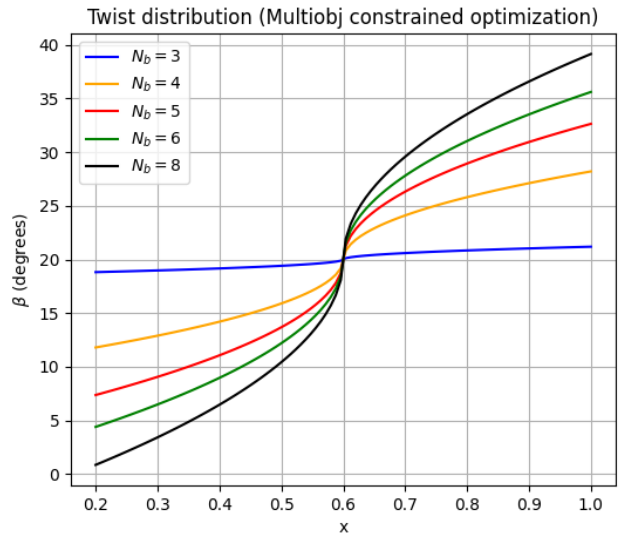
(a) Distribution plot for β vs x for the unconstrained Euler-Lagrangian Objective



(b) Distribution plot for β vs x for the Euler-Lagrangian Objective using Power Constraints

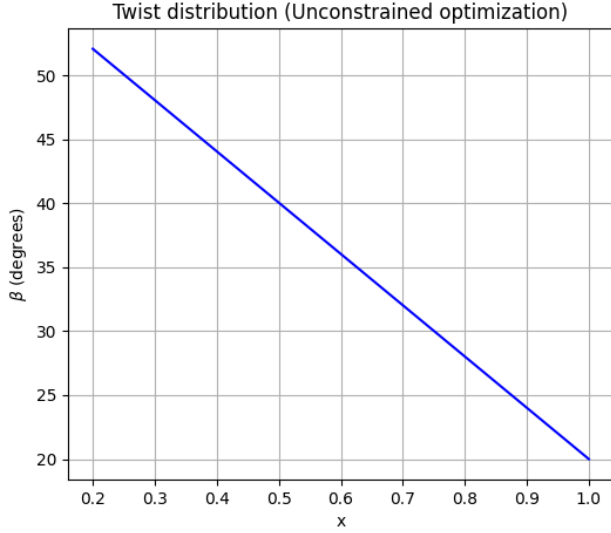


(c) Distribution plot for β vs x for the Multi-objective problem which optimises the Euler-Lagrangian and Sound Pressure Level objectives for thrust and power constraints for $N_b = 3$ blades.

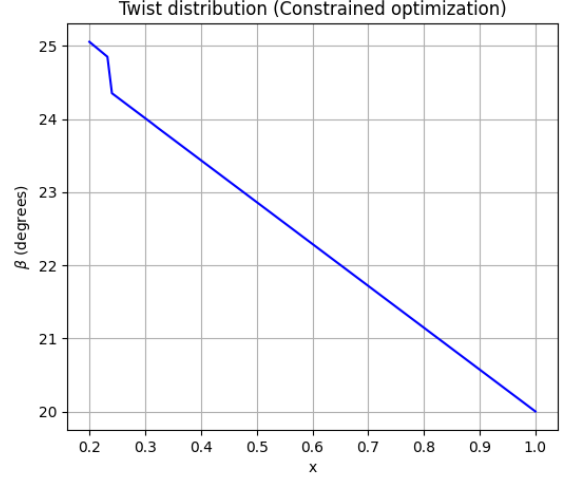


(d) Distribution plot for β vs x for the Multi-objective problem for different number of blades N_b

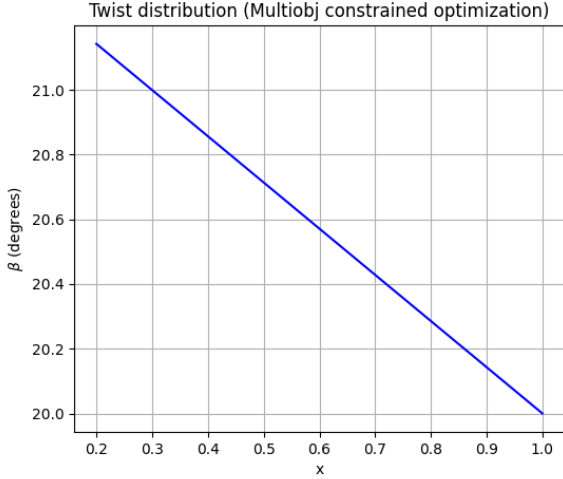
Figure 4: Distribution plots for the twist angle β vs $x = \frac{r}{R}$ for $\mu(x) = \sqrt{|x - 0.6|} \cdot \frac{\text{sign}(x-0.6)}{100}$. Experiments were conducted for different number of blades N_b for the propeller, varying from $N_b = 3$ to $N_b = 8$, which can be observed in **Figure d**.



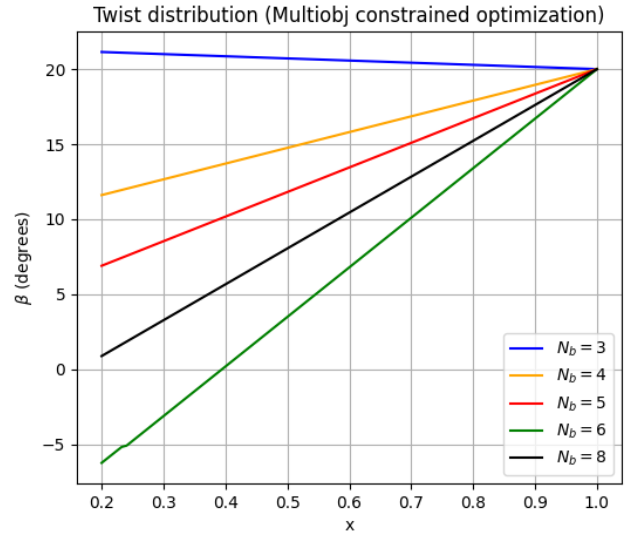
(a) Distribution plot for β vs x for the unconstrained Euler-Lagrangian Objective



(b) Distribution plot for β vs x for the Euler-Lagrangian Objective using Power Constraints



(c) Distribution plot for β vs x for the Multi-objective problem which optimises the Euler-Lagrangian and Sound Pressure Level objectives for thrust and power constraints for $N_b = 3$ blades.



(d) Distribution plot for β vs x for the Multi-objective problem for different number of blades N_b

Figure 5: Distribution plots for the twist angle β vs $x = \frac{r}{R}$ for the simple distribution $\mu(x) = 1 - x$. Experiments were conducted for different number of blades N_b for the propeller, varying from $N_b = 3$ to $N_b = 8$, which can be observed in **Figure d**. It can be observed that for this linear distribution most of the optimal distribution plots obtained are also linear.

6.1 Results of Simplified Problem

A simplified version of the current problem is depicted clearly by the constrained-optimization of the Propeller's Performance Efficiency using the E-L equation. The figures **3(b)**, **4(b)** and **5(b)** depict the resultant twist distributions for maximized η under the power constraint due to the lagrangian.

- The Propeller Efficiency observed at Optimal twist distribution for $\mu(x) = \frac{1}{x-0.5} - 0.5$ is **0.8853052594794984**
- The Propeller Efficiency observed at the optimal twist distribution for $\mu(x) = \sqrt{|x - 0.6|} \cdot \frac{\text{sign}(x-0.6)}{100}$ is **0.43109443028469988**
- The Propeller Efficiency observed at the optimal twist distribution for $\mu(x) = 1 - x$ is **0.879498453052594**

6.2 Justification for Choice of Optimizer

Scipy's SLSQP was implemented as the optimizer for this project. SLSQP is particularly well-suited for problems with both equality and inequality constraints. It can handle nonlinear constraints and is designed to find the minimum of a scalar function subject to equality and inequality constraints. It uses a quadratic approximation to the objective function. Since BFGS is primarily designed for unconstrained optimization, while it can be extended to handle constrained problems using penalty or barrier methods, SLSQP was assumed to be more efficient and straightforward with the constrained problem.

6.3 Optimizer Performance

The optimizer performance was gauged by calculating the time taken by each

6.4 Investigation of the Optimum

The optimal values obtained for **Sound Pressure Level (SPL)** and **Propeller Efficiency (η)** during cruise at **5000 ft** are as follows -

- The optimal value obtained for **Sound Pressure Level (SPL)** is **33.744983080 dB** for the optimal twist distribution for all three distributions of $\mu(x)$ for different values of N_b .
- The Propeller Efficiency observed at Optimal twist distribution for $\mu(x) = \frac{1}{x-0.5} - 0.5$ is **0.8269664679498453**
- The Propeller Efficiency observed at the optimal twist distribution for $\mu(x) = \sqrt{|x - 0.6|} \cdot \frac{\text{sign}(x-0.6)}{100}$ is **0.17884699880944302**
- The Propeller Efficiency observed at the optimal twist distribution for $\mu(x) = 1 - x$ is **0.8201682627052594**

It can be observed from the results above that the selected distribution $\mu(x) = \sqrt{|x - 0.6|} \cdot \frac{\text{sign}(x-0.6)}{100}$ was not optimal as the solver could not achieve sufficient efficiency for the optimal twist angle distribution obtained from it. We can conclude that this does not highlight any faults in the optimization procedure but rather tells us that this is not an optimal distribution assumption for calculating the twist angle. This is a hurdle we came across due to the fact that we did not have readily available distributions for the blade twist angle and had to assume 3 different distributions of $\mu(x)$ to obtain a distribution we could optimize.

6.5 Interpretation of Results

The optimal twist distribution obtained for each mission phase was as follows:

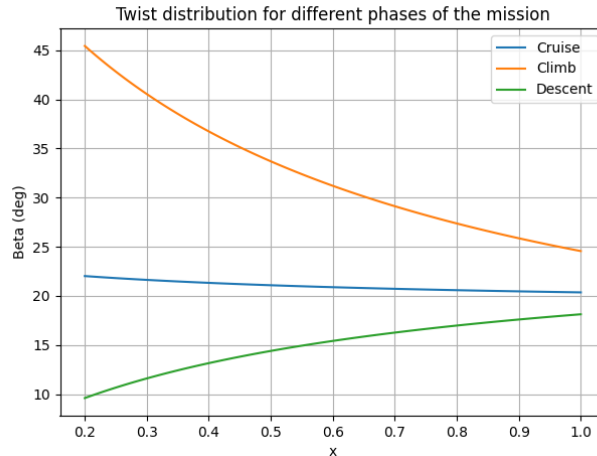


Figure 6: Optimum Twist Distribution along the blade during different mission phases

This can be interpreted by the objectives of our study and how they are affected during each phase of flight for the UAV. During cruise, the drone flies at a constant velocity while trying to maximize the lift generated by the slower moving sections of the blade (near the hub), and reducing the drag generated by the faster moving sections of the blade (near the tip). Thus the resultant optimum twist distribution has a higher twist angle near the hub, that reduces gradually as we

move towards the tip of the blade.

This requirement of increase lift becomes even graver when the vehicle is in its climb phase. The increased thrust requirement results in an exaggerated twist distribution as compared to the cruise phase.

The descent phase is unlike the other two phases, as it does not necessitate the thrust requirement. Due to the thrust constraint not being active anymore, the optimal distribution comes out to be of the opposite fashion.

Furthermore, different values of the Goldstein Tip-Loss Factor (given in Equation eq:goldstein) are tested in order to observe the trend in the twist distribution by changing the lift coefficient of the blades. The results are as follows:

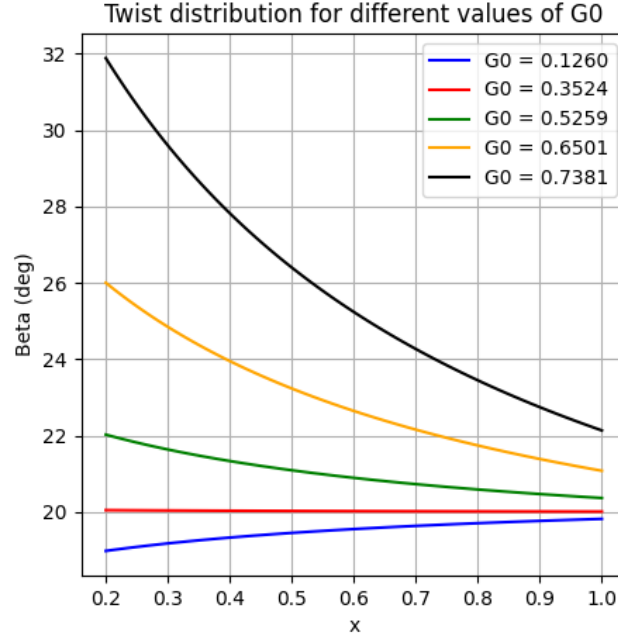


Figure 7: Optimum Twist Distribution along the blade for different values of Goldstein Factor

Next, by altering the value used of λ_L for Equation eq:lagrangian, the value of the Lagrange multiplier changes, hence altering the coefficient of power constraint. The variation is as follows:

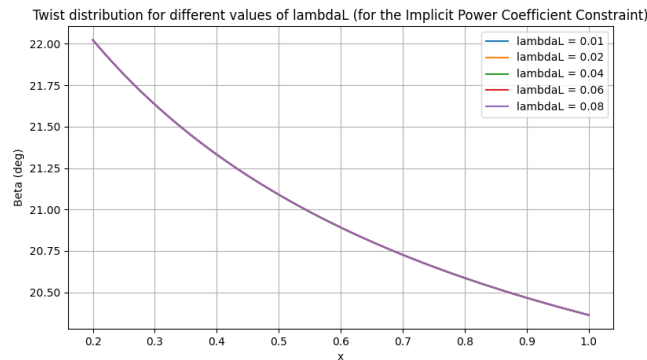


Figure 8: Optimum Twist Distribution along the blade for different values of λ_L (C_P Constraint)

It is observed that the optimum distribution is unaffected by change in the value of λ_L , hence suggesting that the implicit constraint applied on the objective function of η does not serve as an active constraint when it comes to the combined objective.

Finally, the constraint that is explicitly applied to the coefficient of thrust (C_T) is altered to observe the difference

in the twist distribution along the span.

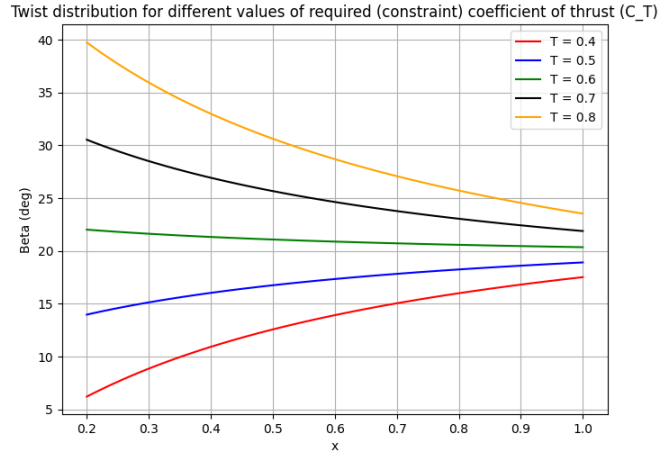


Figure 9: Optimum Twist Distribution along the blade for different constraint values of the Thrust Coefficient (C_T)

It is observed that as the requirement for thrust increases, the twist distribution becomes more and more exaggerated to satisfy the lift requirement. The opposite is seen when the thrust requirement is relaxed, so as to minimize the value of SPL by reducing the twist.

7 Conclusions and Recommendations

References

- [1] Lock, C. N. H., P. R. C. and Fowler, R. G., "Determination of the Optimum Twist of an Airscrew Blade by the Calculus of Variations," *Aeronautical Research Committee, RM No. 2088*, 1942.
- [2] Goldstein, S., "On the Vortex Theory of Screw Propellers," *Proceeding of the Royal Society London, Series A, Vol. 123, No. 792, 1929, pp. 440–465.*, 1929.
- [3] Dorfling, J. and Rokhsaz, K., "Constrained and Unconstrained Propeller Blade Optimization," *Journal of Aircraft*, Vol. 52, No. 4, 2015, pp. 1179–1188.
- [4] R Schlegel, R. K. and Mull, H., "Helicopter Rotor Noise Generation and Propagation," *Technical report, U. S. Army Aviation Materiel Laboratories, Fort Eustis, Virginia*, 1996.